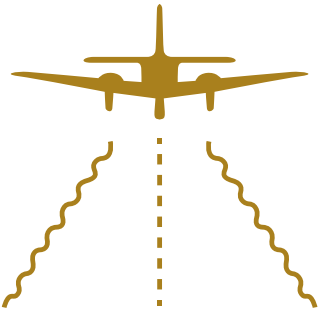
UNOFFICIAL - FOR REFERENCE ONLY

Redhill & Shoreham Aviation



Piper PA28

Checklist

Name: _____

31st March 2021 - Revision 1.0

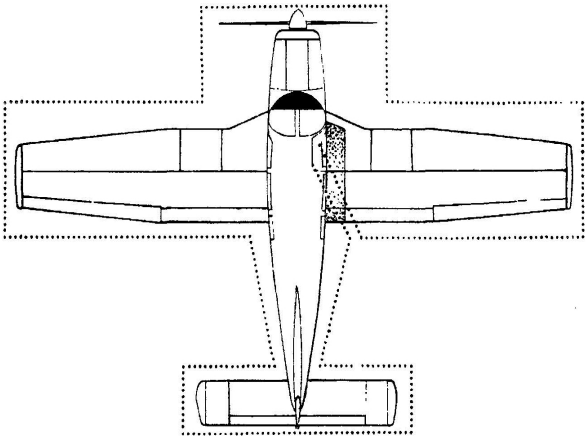
Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

External Checks

Check documents are current and correct. Visually check the aircraft's general condition. Remove tie-down ropes, pitot head cover, chocks and any external control locks. In cold weather, remove even small accumulations of frost, ice or snow from the aircraft.

Check that the aircraft is parked in a free area, so that it may easily move onto the taxyway. Check that a fire extinguisher is available.



1. Cockpit Preparation

Aircraft Documents.....Stowed
Parking Brake.....On
I.F. Screens .....Stowed
First Aid Kit .....Stowed
Fire Extinguisher ..... Checked
Magneto Switches .....Off
Avionics Master Switch .....Off
Circuit Breakers..... Check In
Trim Indicator .....Correct Movement/Set
Flying Controls .....Full Free Correct Movement
Battery Switch .....On
Pitot Heat.....On
Fuel Contents..... Check
Fuel selector.....Emptier Tank
Annunciator/low voltage lights..... Test/On

*Leave cockpit and check externally as necessary, and then re-enter cockpit before carrying out full external pre-flight inspection.

*Recognition/landing lights.....On/Check/Off

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

1. Cockpit Preparation (Continued)

*Anti-collision lights ..... On/Check/Off
*Navigation lights..... On/Check/Off (Leave on at night)
*Pitot heater..... On/Check/Off
*Stall Warner ..... On/Check/Off
*Panel lighting ..... Check (Leave on at night)
Battery switch ..... Off
Pitot static drains ..... Operate
Flaps..... Fully Down/Check All Stages

2. External: Cabin Upper Fuselage

Door..... Closed Flush
Starboard Windows ..... Clean
Upper Fuselage..... Check/Undamaged
Towbar (If Carried)..... Stowed
Baggage Door ..... Secure

3. Starboard Wing

Flap ..... Check/Undamaged
Aileron (Surface, Hinges & Static Wicks) ..... Check/ Undamaged
Wing Tip ..... Undamaged
Strobe & Navigation Light..... Undamaged
Wing Surfaces ..... Undamaged
Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage) ..... Check
Main Gear (Oleo Extension 4½”)..... Check Security
Tyre (Inflation, Creep, No Damage) ..... Check
Brakes (Pads, No Hydraulic Leaks, Undamaged)..... Check
Fresh Air Intake ..... Unobstructed

4. Nose Section

Windscreen ..... Clean
OAT Probe..... Check
Oil Contents (8 Qts Min 2 Qts, Recommended Minimum 6 Qts)..... Check/Cap Secure
Starboard Cowling..... Secure
Propeller/Spinner..... Undamaged/Secure
Air Intakes ..... Unobstructed
Alternator Belt..... Check Tension

Continues on next page

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

4. Nose Section (Continued)

Landing Light ..... Undamaged
Nose Gear (Oleo Extension 3¼") ..... Check Security
Tyre (Inflation, Creep, No Damage) ..... Check
Brake Fluid ..... Check
Port Cowling ..... Secure
Fuel Drain ..... Sample/No Seepage
Port Window ..... Clean

5. Port Wing

Fresh Air Intake ..... Unobstructed
Brakes (Pads, No Hydraulic Leaks, Undamaged) ..... Check
Tyre (Inflation, Creep, No Damage) ..... Check
Main Gear (Oleo Extension 4½") ..... Check Security
Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage) ..... Check
Pitot Head/Static ..... Unobstructed/Secure
Stall Warning Vane ..... Free/Undamaged
Wing Surfaces ..... Undamaged
Wing Tip ..... Undamaged
Strobe & Navigation Light ..... Undamaged
Aileron (Surface, Hinges & Static Wicks) ..... Check/ Undamaged
Flap ..... Check/Undamaged

6. Rear Fuselage, Fin And Stabilator

Radio Aerials ..... Secure
Port Fuselage & Window ..... Undamaged/Clean
Fresh Air Intake ..... Check
Fin Surface ..... Undamaged/Secure
Rudder & Nav Light ..... Undamaged
Stabilator ..... Free/Undamaged
Air Conditioning Scoop (If Fitted) ..... Flush/Undamaged
Starboard Fuselage & Window ..... Undamaged/Clean
Static Vents (If Fitted) ..... Unobstructed/Clean

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Before Engine Start Checks

1.	Flaps	Retracted
2.	External Checks	Completed
3.	Passenger Briefing	Completed
4.	Cabin Door	Closed & Latched
5.	Seats Seat Backs Upright	Adjusted & Locked
6.	Seat Belts & Shoulder Straps	Fitted & Secure
7.	Alternate Static Selector (If Fitted)	Check/Normal
8.	Flight Instruments Standby Compass & Deviation Card	Check Condition
9.	Battery Master Switch	On
10.	Annunciator/Low Voltage Light	Check/On
11.	VAC Off Caption	On
12.	Avionics/Radio Master Switch	Off

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Starting

1.	Anti-Collision Light	On
2.	Fuel Selector	Tank With Lowest Contents
3.	Mixture	Exercise/Set Rich
4.	Throttle	Exercise/Set ¼”
5.	Electric Fuel Pump	On, check pressure, Off
6.	Prime	As Required & Lock
7.	Lookout	Clear All Round, Open Window “CLEAR PROP”
8.	Magnetos	On Both
9.	Starter	Operate & Release as Soon as Engine Starts

After Start

1.	Starter Warning Light	Out, If Not Immediately Turn Magnetos Off
2.	Oil Pressure	Within 30 secs 25 PSI Minimum (If Not, Shut down)
3.	RPM	Set to 1200
4.	Alternator/Ammeter	On & Holding Positive Charge
6.	Low Voltage Light	Out
7.	Suction	Check Registering, VAC off Caption Out
8.	Annunciator Panel Lights	All Out
9.	Fuel Pressure	Check
10.	Avionics/Radio Master Switch	On (If Fitted)
11.	Electric Trim (If Fitted)	Check
12.	Flight Instruments	Check (AI erect, DI set, VSI, Turn Co-ord flag away) HSI/RMI Heading, Standby Compass/Slaving Meter
13.	Avionics	Checked and Set As Required
14.	Auto-Pilot (If Fitted)	Check and Off
15.	Magnetos	Check at 1200 RPM
16.	Radios	Radio Check, ATIS (if available) & Taxi Clearance
17.	Altimeters	Cross Check & Set

Taxying

1.	Brakes & Nosewheel Steering	Check Operation Satisfactory For Both Pilots
2.	Flight Intruments	During Turns Check Compass & D.I turning in same direction, turn Indicator: Turn & Skid. AI steady

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Power Checks

1.	Position	Into wind. Avoid Rough Ground and Loose Stones. Clear of Aircraft and Personnel
2.	Parking Brake	On
3.	Fuel Selector	Change Tank to Fuller Tank
4.	RPM	Check Clear All Round. Set to 2000
5.	Carb. Heat	Check Operation & Return to Cold
6.	Magnetos	Check Each Magneto in Turn. Max Permissible Drop 175 RPM; and Max. Differential is 50 RPM
7.	Alternator Output	Check
8.	Suction	4.9 to 5.1” Hg.
9.	Temperature & Pressure	Check, Steady and Within Limits
10.	RPM	Smoothly Close the Throttle, Check Idle.Temps and Pressures Steady, Within Limits. Then Reset 1200

Pre-Takeoff Checks

1.	Trim	Set for Take-Off
2.	Throttle Friction	Finger Tight
3.	Mixture	Mixture Rich.
4.	Magnetos	Magnetos on Both. Master Switch On
5.	Carb Heat	Off
6.	Fuel Selector	On Fuller Tank, Check Contents
7.	Fuel Pump	On, Pressure Checked
8.	Flaps	Set (Normal 0°)
9.	Engine Instruments	Check (Temperatures & Pressures)
10.	Seat Belts / Seat Backs	Secure/Erect
11.	Cabin Doors	Locked/Latched
12.	Annunciator Panel	Test/All Lights Out
13.	Flying Controls	Full Free Correct Movement
15.	Radio & Transponder	Confirm Radio/Nav Set. Transponder set, Alt.
16.	Flight Instruments	Check/Altimeters Set. Pitot Heat as Required
17.	Lights	Recognition/Landing Lights and Anit-Cols Lights On
18.	Take-Off Brief	Review Speeds, Crosswind, Departure Brief & Emergency Procedures

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Take Off

Normal Take-Off

1.	Lookout	Obtain Clearance, Check Approach Clear Visually
2.	Line up	On Runway Centreline, Ensure Nosewheel Straight, D.I. Indicating Runway Heading. Controls Free
3.	Brakes	Off, Heels on Floor
4.	Throttle	Smoothly Open to Full; Check Power
5.	Oil	Temperature and Pressure Steady and in Green
6.	Airspeed	Increasing
7.	Elevator Control	Lift Nose by 60 Kts
8.	Climb Speed	80 Kts (Clean).

Short Field Take-Off (Hard Surface)

1.	Flaps	25°
2.	Brakes	Hold (Keep Moving During a Soft Field Take-Off)
3.	Throttle	Fully Open, Check power, Oil Temp and Pressure
4.	Brakes	Release
5.	Elevator Control	Slight Back Pressure
6.	Climb Speed	80 Kts (Best angle of Climb is 65 Kts)

After Take-Off

1.	Lookout	Point Selected Left of Nose, Check Drift
2.	Flaps	Retract Above 200’ a.g.l, Not Below 65 Kts
3.	Oil	Temperature and Pressure Steady, Within Limits
4.	Recognition/Landing Lights	Off
5.	Fuel Pump	Off (Above 1000ft a.g.l)

Climb

Normal Climb

1.	Airspeed	Clean: 80 Kts (Best Rate of Climb)
2.	Throttle	Fully Open
3.	Mixture	Rich Below 3000’

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

En-Route Climb		
1.	Airspeed	90 Kts
2.	Throttle	Fully Open
3.	Mixture	As Required

En-Route Checks (FREDA)		
1.	**F**uel	Check Contents/Balanced
2.	**R**adio / Radio aids	Check Volume; Frequency Change if Required / Radio Aids Identified
3.	**E**ngine	Oil Temperature and Pressure Steady and Within Limits. Ammeter Charging. Suction Within Limits. Check for Carb. Icing. Mixture as Required.
4.	**D**irection Indicator	Check D.I.
5.	**A**ltimeter	Set

HASELL Checks (Stalls)		
1.	**H**eight	Sufficient to Recover by 2000' AGL
2.	**A**irframe	Flaps as Required. Brakes off
3.	**S**ecurity	Doors, Windows and Harnesses Secure; No Loose Articles
4.	**E**ngine	Check Oil Temperature and Pressure, Mixture as Required, Check Fuel Contents, Fuel Pump On, Clear Carb Ice
5.	**L**ocation	Clear of Airfields, Built Up Areas, Controlled Airspace or Cloud and Danger Areas
6.	**L**ookout	Check No Aircraft Around, Particularly Below

HELL (For Continued Exercise)		
1.	**H**eight	Sufficient to Recover by 2000' AGL
2.	**E**ngine	Check Oil Temperature and Pressure, Mixture as Required, Check Fuel Contents, Clear Carb Ice
3.	**L**ocation	Clear of Airfields, Built Up Areas, Controlled Airspace or Cloud and Danger Areas
4.	**L**ookout	Check No Aircraft Around, Particularly Below

Note: Reset D.I after completion of the exercise

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Airfield Approach Checks: FFREDA

1.	**F**uel	Check Quantity/Balanced
2.	**F**uel Pump	On
2.	**R**adio	Correct Frequency Volume Up. Radio Aids Identified
3.	**E**ngine	Temperature and Pressure in Limits, Mixture Rich
4.	**D**irection Indicator	D.I. Reset
5.	**A**ltimeters	Altimeters Set

Pre-Landing Checks

1.	**B**rakes	Brakes Off; Pressure Up
2.	**M**ixture	Mixture Rich
3.	**C**arb. Heat	Carb. Heat, Check for Icing and then Return to Cold
4.	**F**uel	Fuel On, Sufficient for Go Around
5.	**O**il	Oil Temperature and Pressure Within Limits
6.	**H**atches and Harnesses	Pilot and Passenger Secure
7.	**L**ights	Recognition/Landing Lights On

Approach and Landing

Normal Landing

1.	Carb. Heat	Apply Full Heat Before Closing Throttle
2.	Flaps	Select 25° When Speed is Below V_{FE}
3.	Airspeed	Initially 70 Kts. Final Approach, 63 Kts, Flap 40°
4.	Touchdown	Mainwheels First, Lower Nose Wheel Gently
5.	Braking	Minimum Required

Flapless Landing

1.	Airspeed	75 Kts

Shortfield Landing

1.	Flaps	40°
2.	Airspeed	65 Kts
3.	Touchdown	Main Wheels, Lower the Nose
4.	Brakes	Apply Positively

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Glide Approaches

Airspeed	75 Kts
Flap	As Required

Notes:
1. During prolonged glides; the engine should be warmed every 500'.
2. The approach speeds are for calm conditions, in gusty or turbulent conditions the figures should be increased.

Go-Around (Missed Approach)

1.	Throttle	Full Power
2.	Carb. Heat	Cold
3.	Airspeed	65 Kts Until Clear of Obstacles, then 80 Kts
4.	Flaps	If 30°, Retract to 25° as Soon as Possible. Retract Remaining 25° Above 200'
5.	Radio	Inform ATC

Continues on next page

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

After Landing

Clear the runway and stop the aircraft

1.	Carb Heat	Cold
2.	Flaps	Up
3.	Fuel Pump	Off
3.	Pitot heat	Off
4.	Strobe Lights / Landing Lights	Off / As Required
5.	Trimmers	Set Neutral
5.	Transponder	Standby
6.	Non-essential Electrics	Off

Shut Down

1.	Park	Into Wind, Nosewheel Straight
2.	Parking brake	On
3.	Engine	1200 RPM for 30 secs; Check Oil Temperature and Pressure
4.	Magnetos	Check
5.	Avionics/Radio Master Switch	Off
6.	Mixture	Full Lean
7.	Throttle	Closed

After Engine Stop

1.	Magnetos	Off, Key Out
2.	Exterior Lights	All Off
3.	Fuel Selector	Off
4.	Interior Lights	Off
5.	Battery & Alternator Masters	Off
6.	Harnesses	Tidy
7.	External	Pitot Cover On. Lock up and tie down. Use Chocks if necessary

End of Normal Checklist

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Emergencies

Engine Failure After Takeoff

If the engine fails after take-off you should (as soon as you notice the loss of power):

1. Lower the Nose and Maintain the Aircraft's Gliding Speed
2. Close the Throttle
3. Select the Most Suitable Landing Area, Compatible With Obstructions and Wind Direction; Preferably Straight Ahead or Within 30° Either Side
4. Maintain Speed and Use Flap as Necessary
5. If Time Permits: - Mixture to I.C.O., Fuel Off, Fuel Pump Off, Magnetos Off, Door Unlocked and Harnesses Tight, Mayday Call, Master Switch Off
6. After Touchdown Leave the Aircraft as Quickly as Possible

Crash Drill

1. Mixture I.C.O, Fuel Pump Off
2. Fuel Selector Off
3. Magnetos Off
4. Brakes Off
5. Harnesses Very Tight
6. Hatches Unlatched
7. Master Switch Off (After Electrical Services are No Longer Required)

Continues on next page

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Engine Fire During Start

1. Starter - Crank Engine
2. Mixture - Idle Cut Off
3. Throttle - Open
4. Electric Fuel Pump - Off
5. Fuel Selector - Off
6. Abandon Aircraft if Fire Continues

Engine Fire In The Air

1. Fuel Selector - Off
2. Throttle - Closed
3. Mixture - Idle Cut-Off
4. Electric Fuel Pump - Check Off
5. Heater and Defroster - Off
6. Proceed with Power Off Landing Procedure

Electrical Fire In The Air

1. Master Switch Off, Alternator Off
2. Vents - Open
3. Cabin Heat - Off
4. Land as soon as Practicable.

Cabin Fire In The Air

1. Master Switch Off
2. Vents, Cabin Air/Heat Closed
3. Use Fire Extinguisher and Then Ventilate The Cabin
4. Land as Soon as Possible.

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Forced Landing Without Power

1. Adopt Glide and Trim for Correct Gliding Speed
2. Check Wind Direction
3. Select Landing Area and 1000' Area
4. Plan Circuit
5. Complete Failure Checks:
    - Switch Tanks
    - Electric Fuel Pump On
    - Mixture Rich
    - Carb. Heat Change Position
    - Magnetos on Both
    - Primer Locked
6. Mayday Call on Radio
7. Complete Crash Drill
8. Use Full Flap for the Touchdown if Possible
9. Apply Heavy Braking

Forced Landing With Power

1. Assess the Time Available and Check Fuel State, Hours of Daylight and Rate of Weather Deterioration
2. Check Wind Direction
3. Select Suitable Landing Area
4. Carry Out an Inspection Run at Slow Safe Cruising Speed; Initially at 500' to Check for Obstructions and Secondly at 100' Over the Landing Area to Reassess the Approach Path, Landing Surface and Overshoot Area (if time permits)
5. Set up For a Short or Soft Field Landing (as appropriate)

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Ditching

1. Set up Minimum Rate of Descent (75 Kts)
2. Mayday Call
3. Secure or Jettison Heavy Objects
4. With High Winds and Heavy Seas: Approach Into Wind.
   With Light Winds and Heavy Swells: Approach Parallel to Swells.
5. Unlatch the Door
6. Touchdown in a Level Attitude
7. Evacuate Through the Cabin Door; if Necessary Open Window to Equalise Pressure

Notes:

1. Ideally, land ahead and either side of a small ship.
2. **Do not inflate life jackets in the cockpit.**

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Low-Voltage/Electrical Failure

1. Check Ammeter, if Reading Zero, Alternator Off
2. Reset Circuit Breaker
3. Alternator On
4. Reduce Electrical Load

Notes:

1. If total electrical failure occurs, keep an extra good lookout. Land when safe to do so, clear the runway as quickly as possible and phone ATC at once.

Radio Failure

1. Check Correct Frequency
2. Check Squelch Setting (if applicable)
3. Check Volume
4. Check Microphone Plug Secure
5. Check Circuit Breaker
6. Try to Communicate Using the Other PTT Switch

Notes:

1. If the radio is unserviceable: flash the landing light on approach and land on visual signals.
2. If flying outside of the circuit, follow lost comm procedure from AIP.

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Uncertain of Position / Lost Procedure

Uncertain of Position

Inform ATC immediately on the frequency you are in communication with.

Check flight plan, log, compass and heading for gross errors; calculate DR position; from flight time decide radius of circle of uncertainty; mark on map using DR position as Centre; estimate new heading from DR position to regain destination or track. Utilise Nav aids including moving map if available.

Lost Procedure

a) Radio Available

Check frequency selection; declare that you are lost; make maximum use of aids available; consider increasing height to improve range (taking account of controlled airspace). R/T procedure: urgency or distress call to obtain fix. You are no longer lost. Follow the ATC instructions in full.

b) Radio Not Available

i. Check remaining fuel and calculate endurance; consider daylight remaining, weather tendency, nature of terrain in relation to probable position, minimum safe altitude; re-check flight log for errors; switch transponder, if available, to emergency code.

ii. Map reading from ground to map and try to identify the likely area from the nature of the terrain before checking other features. Look for prominent features in the distance as well as nearby; adhere to a calculated heading and keep map properly orientated.

iii. Decide on a positive course of action; relative merits of square search, flight towards prominent or line feature outside of circle of uncertainty; consider precautionary landing (see Ex 17) if you are short of fuel or are unable to continue the flight.

End of Emergency Checklist

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Altimeter Setting Procedures

		No. 1	No. 2
Pre-flight	Check +50 ft / -75 ft of datum, 50 ft of each other	Airport QFE Airport QNH	Airport QFE Airport QNH
Departure	For take-off Cleared to a FL.	Airport QNH 1013	Airport / zone QNH Airport / zone QNH
Cruise	Above Transition Alt At / below Transition Alt	1013 Regional QNH	Regional QNH Regional QNH
Note	In or below TMA use TMA QNH not regional QNH		
Descent	Cleared to FL Cleared to Altitude	1013 Zone / airport QNH	Zone / regional QNH Zone / regional QNH
Final Approach / Circuits		Airport QNH / QFE	Airport QNH
Missed Approach		Airport QNH	Airport QNH
Note	Airport QNH is valid within 25m of airport		

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.

UNOFFICIAL - FOR REFERENCE ONLY

Recommended Operating Speeds / Limitations

		Archer 181	Warrior 161
Take-Off	Rotate (depending on weight)	60	60
	Reduced by 5 kts for soft fields		
Climb	0° Flap / Best Rate	79	79
	0° Flap / Best Angle	65	63
	Cruise Climb	90	90
	Minimum speed of 60 kts for flap retraction		
Cruise	Normal / 2200 RPM	100	90
	Bad Vis. 20° Flap / 2000 RPM	70	70
Approach	Normal (25° Flap)	70	70
	Glide	75	75
	Flapless	75	75
	Performance / Short-field (40° Flap)	65	65
	1. *For max performance approach speed is 63 kts*		
	2. *Add 5kts to take-off speeds and V_{AT} in strong winds / turbulence, reducing flap settings as necessary*		
Max Airspeeds	V_{FE} (White Arc)	102	103
	V_{NO} (Green Arc)	125	126
	V_{NE} (Red Line)	154	160
	V_A (Manoeuvring Speed @ MAUW)	113	111
Max X-Wind		17	17

Crosswind Angle	10°	20°	30°	40°	50°	60°	70°	80°	90°
Crosswind Factor	0.2	0.3	0.5	0.6	0.7	0.8	0.9	0.9	1.0

Not an approved checklist. The aircraft POH/AFM and operator procedures are authoritative.